CS1 Week 9
Empty

Frequency Response, Bode Plots

Recap Last Week

Quiz 1

What influences the steady state error?

A. the number of poles & zeros

B. the number of integrators

C. the Bode gain

D. the reference value

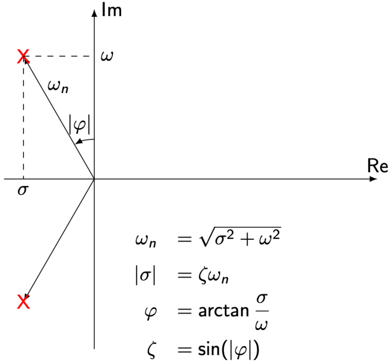
Quiz 2

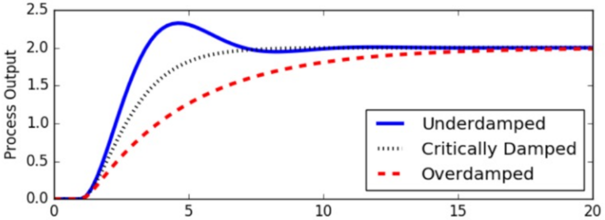
What time-domain specifications did we introduce for 2nd order systems?

- **A.** Settling time
- **B.** Peak time
- **C.** Tea time
- **D.** Stabilty transformation
- **E.** Critical time
- **F.** Peak Overshoot
- **G.** Rise time
- **H.** Peak Overshhot ratio

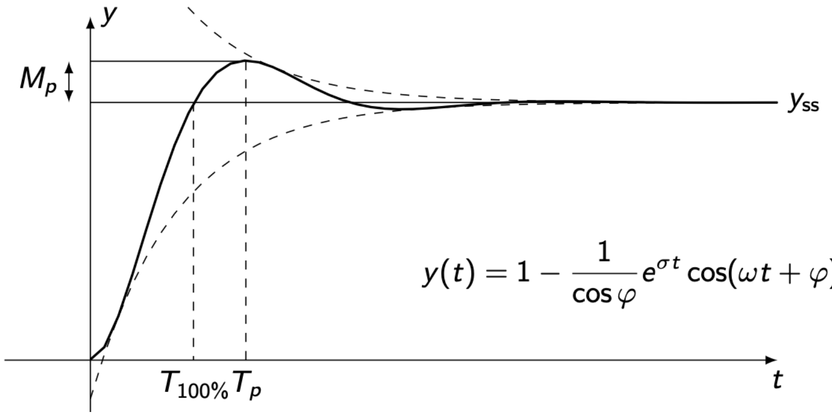
Step Response of a 2nd Order System

$$G(s) = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n + \omega_n^2} \quad \Leftrightarrow \quad \begin{cases} \dot{x} = \begin{bmatrix} 0 & 1 \\ -\omega_n^2 & -2\zeta\omega_n \end{bmatrix} x + \begin{bmatrix} 0 \\ \omega_n^2 \end{bmatrix} u \\ y = x; \end{cases}$$





For an underdamped system ($\zeta < 1$):



$$y(t) = 1 - \frac{1}{\cos\varphi} e^{\sigma t} \cos(\omega t + \varphi), \quad t \geq 0.$$

Specifications:

Settling Time: $T_d = \frac{1}{|\sigma|} \log\left(\frac{100}{d}\right)$

Peak Time: $T_p = \frac{\pi}{\omega}$

Peak Overshoot: $M_p = e^{\frac{\sigma\pi}{\omega}}$

Peak Overshoot Ratio: $\zeta^2 = \frac{\ln(M_p)^2}{\pi^2 + \ln(M_p)^2}$

Rise Time: $T_{100\%} = \frac{\frac{\pi}{2} - \varphi}{\omega} \approx \frac{\pi}{2\omega_n}$

Quiz 3

Approximate the system $G(s) = \frac{1}{4} \cdot \frac{1}{(s+10)(s+1+j)(s+1-j)}$ to 2^{nd} order

Course Schedule

	Subject	Week
Modeling	Introduction, Control Architectures, Motivation	1
	Modeling, Model examples	2
	System properties, Linearization	3
Analysis	Analysis: Time response, Stability	4
	Transfer functions 1: Definition and properties	5
	Transfer functions 2: Poles and Zeros	6
	Proportional feedback control, Root Locus	7
	Time-Domain specifications, PID control	8
	Frequency response, Bode plots	9
	The Nyquist condition, Time delays	10
Synthesis	Frequency-domain Specifications, Dynamic Compensation, Loop Shaping	11
	Time delays, Successive loop closure, Nonlinearities	12
	Describing functions	13
	Intro to Uncertainty and Robustness	14

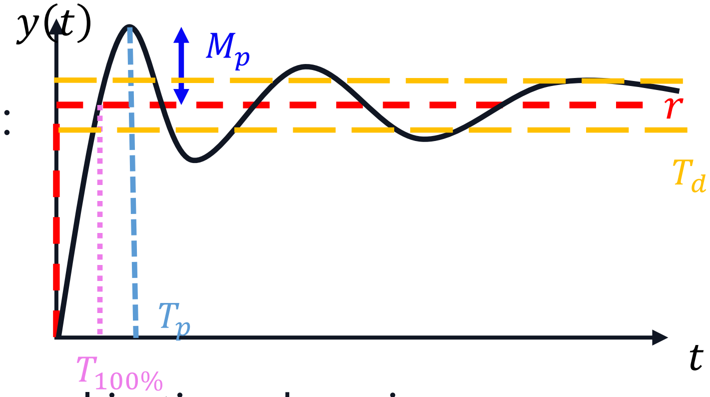
Today

1. Frequency Response Plots
2. Bode Plots
3. Polar Plots

1. Frequency Response Plots

Introduction

- Last week we look at time-domain specifications:



- This means we analyzed the response to a test signal in time domain

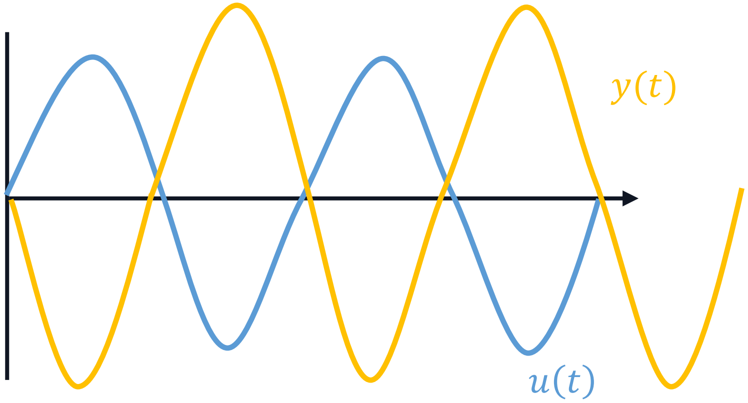
$\longrightarrow$ **Signal behavior over time**

- But we said inputs are often sinusoids. What if we want to look at how a system responds to different frequencies?

$\longrightarrow$ **Signal behavior over frequency**

Why frequencies?

- Many signals can be written as complex exponentials: $u(t) = e^{(j\omega t)} = cos(\omega t) + jsin(\omega t)$
- And with a stable system we have: $y_{ss}(t) = G(j\omega)e^{(j\omega t)} = \underbrace{|G(j\omega)|}_{\text{amplification}}e^{\left(j\left(\omega t + \underbrace{\angle G(j\omega)}_{\text{phase shift}}\right)\right)}$
- Input signals change amplitude and phase



Because of this…

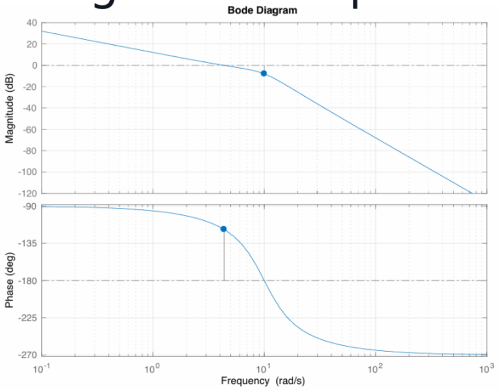
In the extreme case of amplification, resonance can occur

Frequency Response Plots

- Idea: Plot $G(j\omega)$ for different frequencies ω. Remember phase change $\angle G(j\omega)$ and magnitude change $|G(j\omega)|$ are both dependent on ω
- 2 Options:

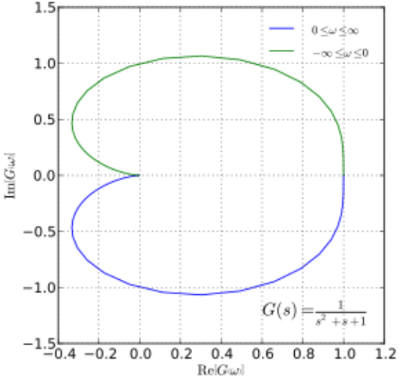
Bode Plot

Two separate plots for magnitude and phase



Polar/Nyquist Plot

A parametric curve showing $G(j\omega)$ in the complex plane, where ω is implicit



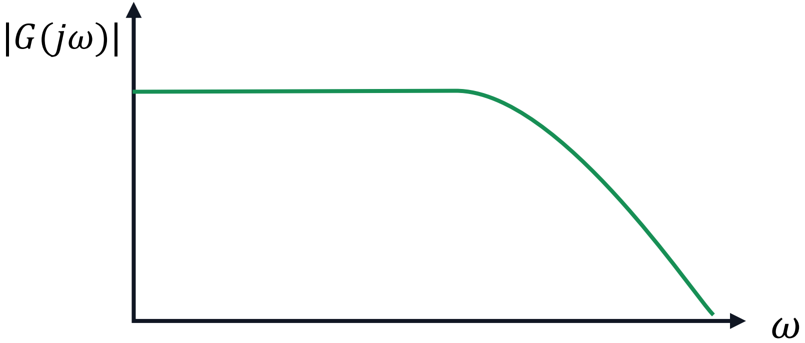
2. Bode Plots

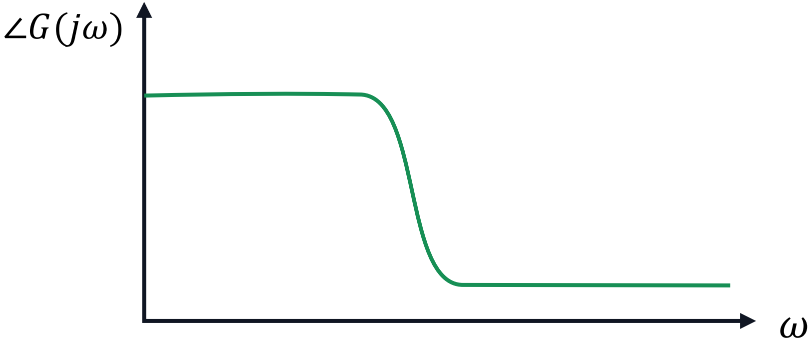
Bode Plot

- Important: For a given input with frequency ω, an LTI-system will have constant:
  - $|G(j\omega)|$
  - $\angle G(j\omega)$

- The frequency of the output always stays the same ω

- This is exactly what the Bode plot shows: magnitude & phase as a function of ω
- The Bode plot therefore consists of two graphs as such:





Bode Plot

- Horizontal Axis: Frequency ω plotted on a $\log_{10}$ scale
- Vertical Axis:
  - Magnitude $|G(j\omega)|$ plotted in deciBels (dB)
  - Phase plot $\angle G(j\omega)$ in degrees

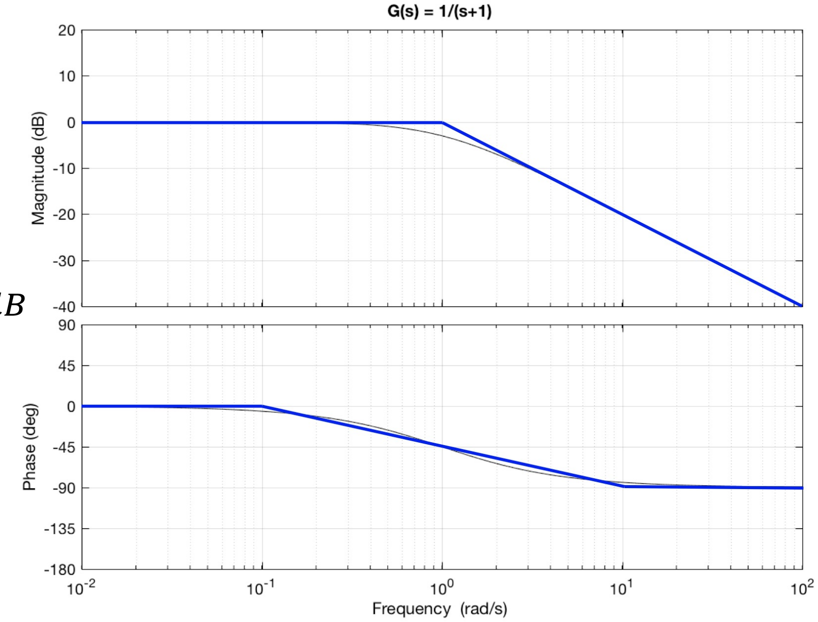
$$|G(j\omega)|_{dB} = 20 \cdot \log_{10}|G(j\omega)|$$

$$|G_1(j\omega)| \cdot |G_2(j\omega)| = |G_1(j\omega)|_{dB} + |G_2(j\omega)|_{dB}$$

$$\angle G_1(j\omega)G_2(j\omega) = \angle G_1(j\omega) + \angle G_2(j\omega)$$

- Inverting a TF is equivalent to a reflection about the horizontal axis

$$|G(j\omega)^{-1}|_{dB} = -|G(j\omega)|_{dB}$$



$$|G(j\omega)|_{dB} = 20 \cdot \log_{10}|G(j\omega)|$$

Example

- You are given the following system: $G(j\omega) = \frac{1}{j\omega+10}$

Evaluating the *magnitude*:

$$|G(j\omega)|_{[dB]} = 20 \cdot \log_{10}\left(\frac{1}{|j\omega + 10|}\right)$$

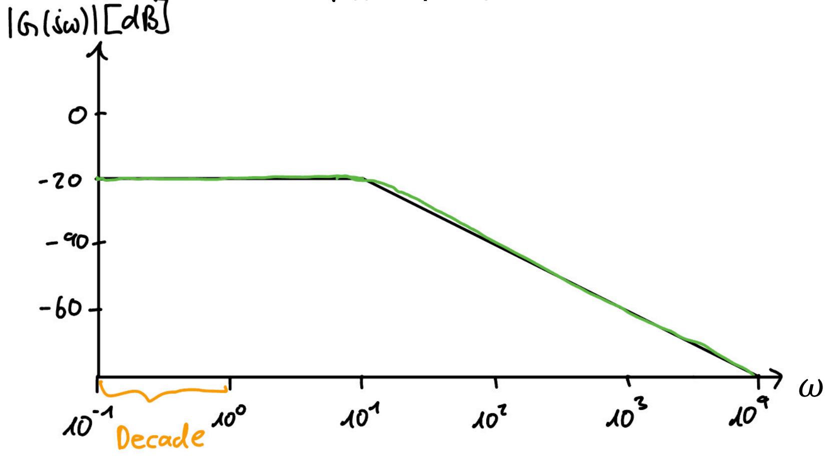
$$= 20 \cdot \log_{10}\left(\frac{1}{\sqrt{\omega^2 + 10^2}}\right)$$

Evaluating the *phase*:

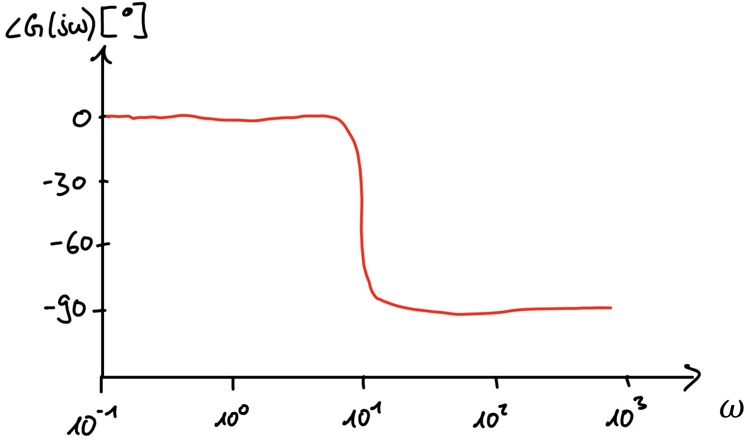
$$\angle G(j\omega) = \angle \frac{1}{j\omega+10} = \angle 1 - \angle(j\omega + 10)$$

$$= -\arctan\left(\frac{\omega}{10}\right)$$

plot:



plot:



Bode Form:

$$G(s) = \frac{k_{bode}}{s^q} \frac{\left(\frac{s}{-z_1}+1\right)\left(\frac{s}{-z_2}+1\right)\dots\left(\frac{s}{-z_m}+1\right)}{\left(\frac{s}{-p_1}+1\right)\left(\frac{s}{-p_2}+1\right)\dots\left(\frac{s}{-p_{n-q}}+1\right)}$$

Sketch Bode Plots

- So how do we sketch these plots by hand?
- Suppose you are given a system $G(s) = \frac{(s-100)}{(s+10)(s-1000)}$

1. Bring TF into *Bode Form* and read out bode gain, p_i & z_i:

2. sort p_i & z_i from smallest to largest (absolute value!)

3. We follow the rules and change our steepness whenever we "encounter" a pole or zero:

Phase \ Magnitude	−20 dB/dec	+20 dB/dec
−90°	stable pole	non-minimum phase zero
+90°	unstable pole	minimum phase zero

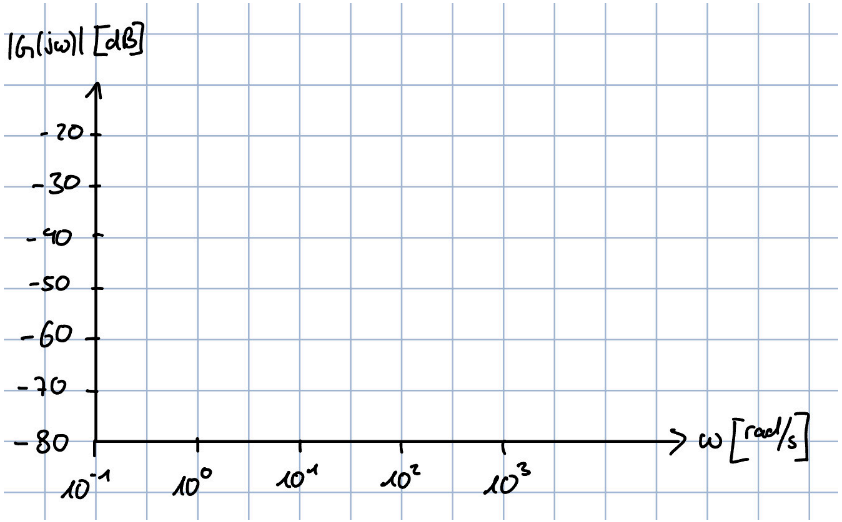
The phase change happens across two decades, one before and one after ω!

Phase \ Magnitude	−20 dB/dec	+20 dB/dec
−90°	stable pole	non-minimum phase zero
+90°	unstable pole	minimum phase zero

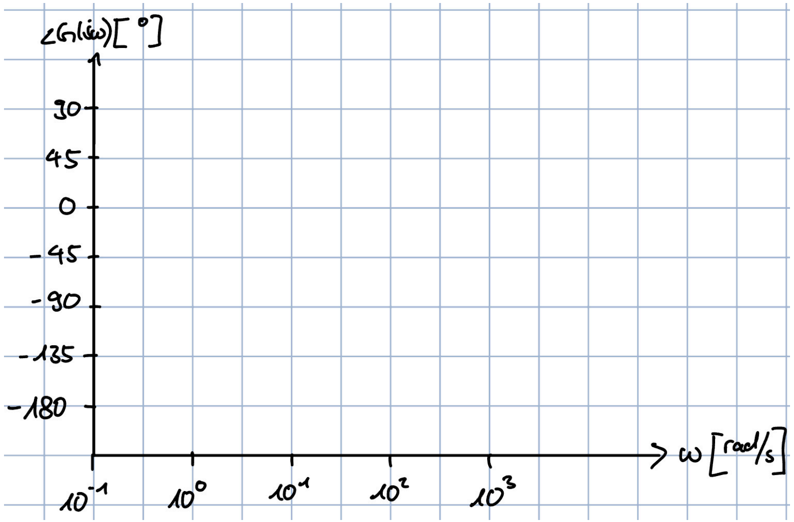
Sketch Bode Plots

$p_1 = -10, z_2 = 100, p_2 = 1000$

Magnitude Plot:



Phase Plot:



Magnitude / Phase	-20 dB/dec	$+20$ dB/dec
$-90°$	stable pole	non-minimum phase zero
$+90°$	unstable pole	minimum phase zero

More Rules

Special rules:

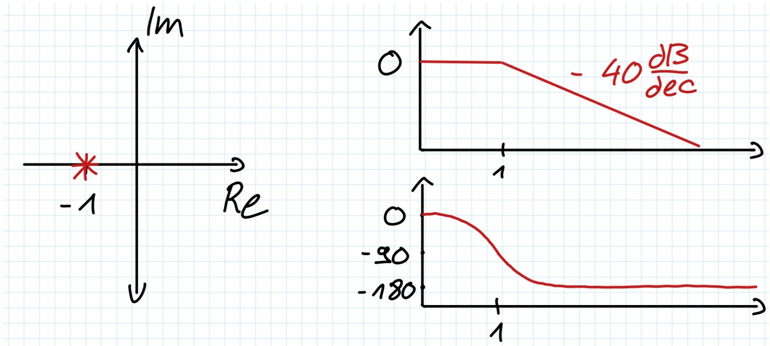
- If TF has pole $p = 0$ (integrator), the plot will start with $-20dB/dec$ and $-90°$
- If TF has zero $z = 0$ (differentiator), the plot will start with $+20dB/dec$ and $+90°$

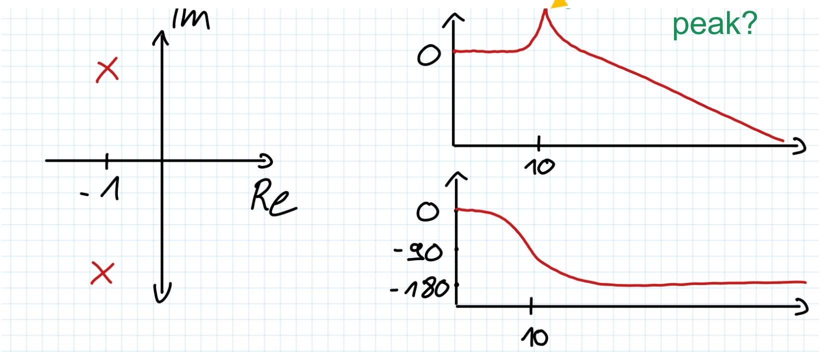
What if we have complex poles/zeros?
Let's compare the following two systems:

$$G_1(s) = \frac{1}{(s+1)(s+1)}$$

$$G_2(s) = \frac{1}{(s+1+10j)(s+1-10j)}$$

Resonance!
How can we characterize this peak?



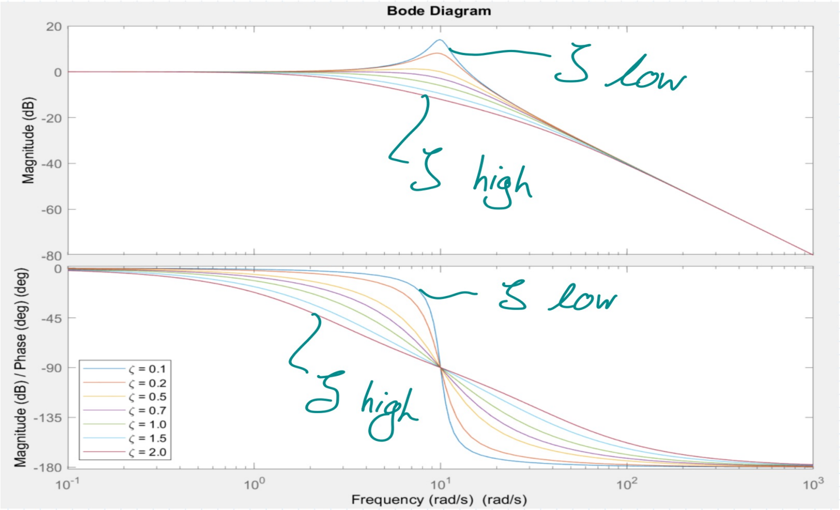


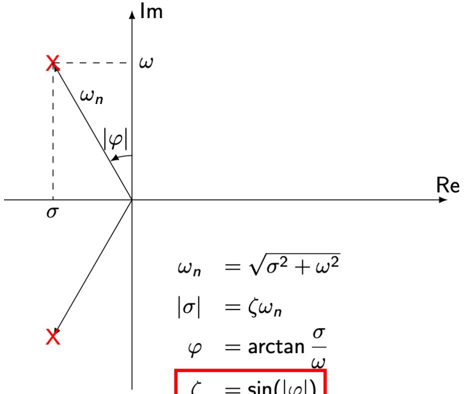
Second Order Systems

Remember how we wrote our second order system: $G(s) = \dfrac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$

- It turns out the dampening factor ζ defines the intensity of the resonance!

$\longrightarrow$ the stronger the dampening, the smaller the resonance!





$\omega_n = \sqrt{\sigma^2 + \omega^2}$

$|\sigma| = \zeta\omega_n$

$\varphi = \arctan\dfrac{\sigma}{\omega}$

$\boxed{\zeta = \sin(|\varphi|)}$

Normalize the system → break frequency is ω_n:

$$G(s) = \frac{1}{\left(\frac{s}{\omega_n}\right)^2 + 2\zeta\left(\frac{s}{\omega_n}\right) + 1}$$

at which frequency we encounter the complex pole

$$|G(j\omega)|_{dB} = 20 \cdot \log_{10}|G(j\omega)|$$

$$|G_1(j\omega)| \cdot |G_2(j\omega)| = |G_1(j\omega)|_{dB} + |G_2(j\omega)|_{dB}$$

$$\angle G_1(j\omega)G_2(j\omega) = \angle G_1(j\omega) + \angle G_2(j\omega)$$

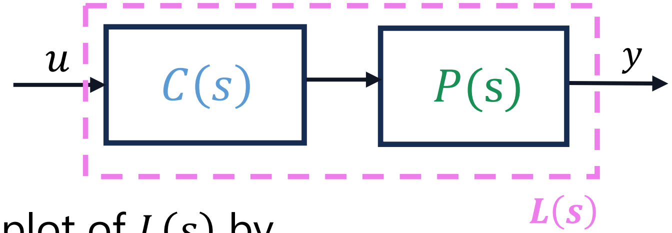
Multiple Systems

- Given two systems, controller and plant
- Magnitudes add up in the Bode plot:

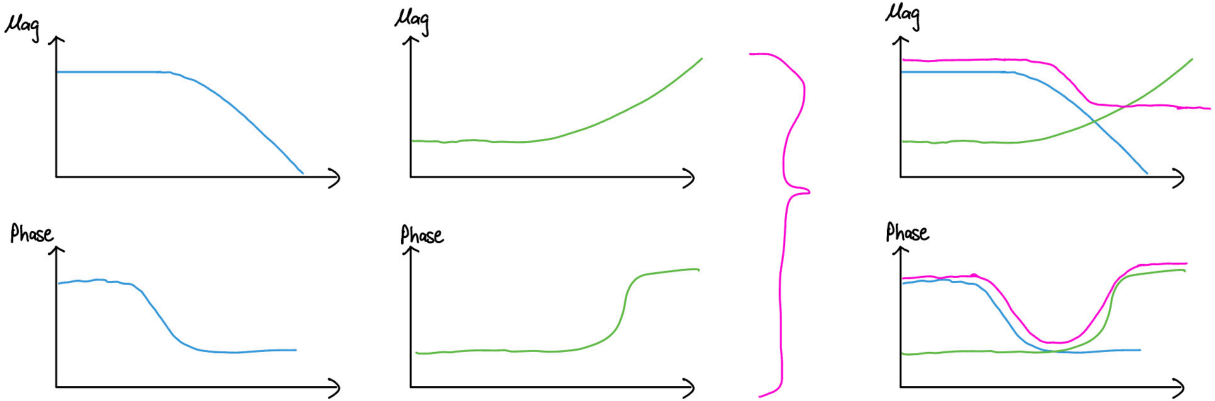
$$|L(j\omega)| = |C(j\omega)| \cdot |P(j\omega)| = |C(j\omega)|_{dB} + |P(j\omega)|_{dB}$$

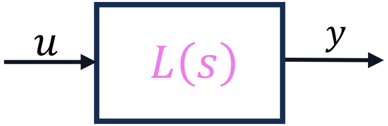
- Phases add up as well:

$$\angle L(j\omega) = \angle C(j\omega)P(j\omega) = \angle C(j\omega) + \angle P(j\omega)$$



if you have the Bode plots of $C(s)$ & $P(s)$, you can find the plot of $L(s)$ by adding them on top of each other!





Bode Plot: Overview

Standard first-order forms:
- Zero: $G(s) = 1 + \tau s$
- Pole: $G(s) \frac{1}{1+\tau s}$

Element type	Formally	Magnitude	Phase
single stable pole	$Re(p_i) \leq 0$	$-20 \frac{dB}{dec}$	$-90°$
single unstable pole	$Re(p_i) > 0$	$-20 \frac{dB}{dec}$	$+90°$
cmplx stable pole	$Re(p_i) \leq 0$	$-40 \frac{dB}{dec}$	$-180°$
cmplx unstable pole	$Re(p_i) > 0$	$-40 \frac{dB}{dec}$	$+180°$
single mnm phs zero	$Re(z_i) \leq 0$	$+20 \frac{dB}{dec}$	$+90°$
single non-mnm phs zero	$Re(z_i) > 0$	$+20 \frac{dB}{dec}$	$-90°$
cmplx mnm phs zero	$Re(z_i) \leq 0$	$+40 \frac{dB}{dec}$	$+180°$
cmplx non-mnm phs zero	$Re(z_i) > 0$	$+40 \frac{dB}{dec}$	$-180°$
positive constant	$k \geq 0$	$0 \frac{dB}{dec}$	$0°$
negative constant	$k < 0$	$0 \frac{dB}{dec}$	$\pm 180°$
differentiator	$\tau \cdot s$	$+20 \frac{dB}{dec}$	$+90°$
integrator	$\frac{1}{\tau \cdot s}$	$-20 \frac{dB}{dec}$	$-90°$
time delay	$e^{s \cdot \tau}$	$0 \frac{dB}{dec}$	$-\omega \cdot \tau$

1. Bring $G(s)$ into BODE form, calculate poles and zeroes and draw them in the bode diagrams (magnitude and phase).

$$G(s) = \frac{k_{Bode}}{s^q} \frac{(\frac{1}{-z_1}+1)(\frac{1}{-z_2}+1)\ldots(\frac{1}{-z_m}+1)}{(\frac{1}{-p_1}+1)(\frac{1}{-p_2}+1)\ldots(\frac{1}{-p_{n-q}}+1)}$$

- The magnitude slope changes at the break frequency $\omega = \frac{1}{\tau}$
- The phase change happens across two decades, one before and one after ω
- Complex conjugate elements double the magnitude slope and phase changes
- Bode Plots are additive

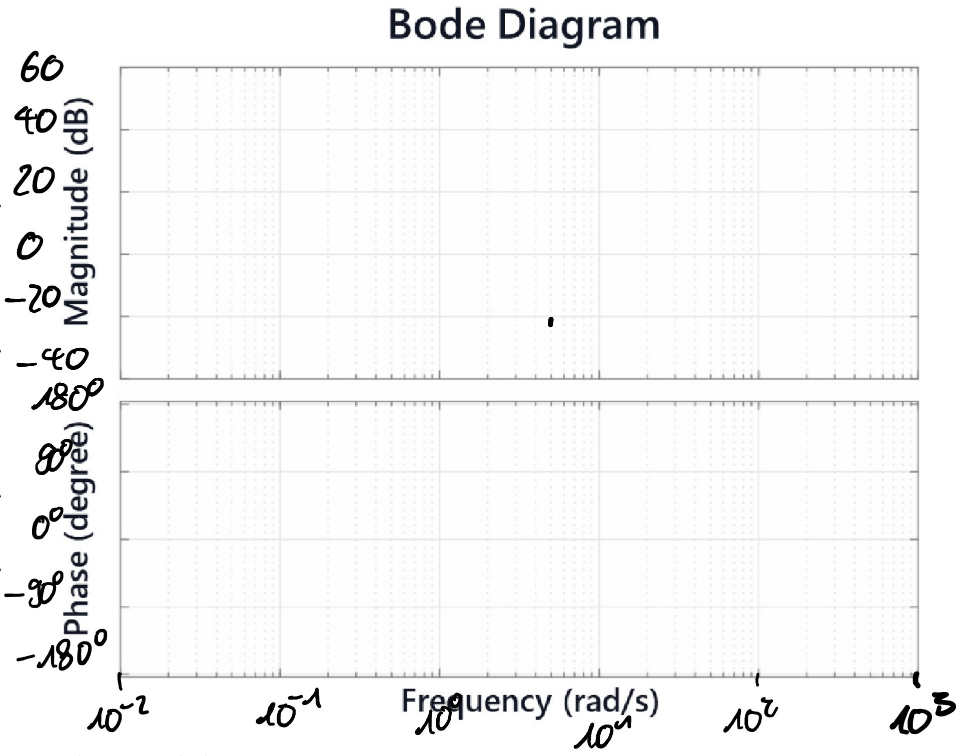
2. Find starting point at $\omega = 0$. If there are no integrators $\frac{1}{s}$ or differentiators s compute the dB value of k_{bode} and draw a horizontal line. Phase graph in this case starts at $0°$ if $k_{bode} > 0$ or at $-180°$ if $k_{bode} < 0$.
   Integrator $\frac{1}{s}$: Slope is instantly $-20\frac{dB}{dec}$. Phase starts at $-90°$
   Differentiator s: Slope is instantly $+20\frac{dB}{dec}$. Phase starts at $+90°$

3. Draw the rough graphs

Constructing Bode Plot from TF

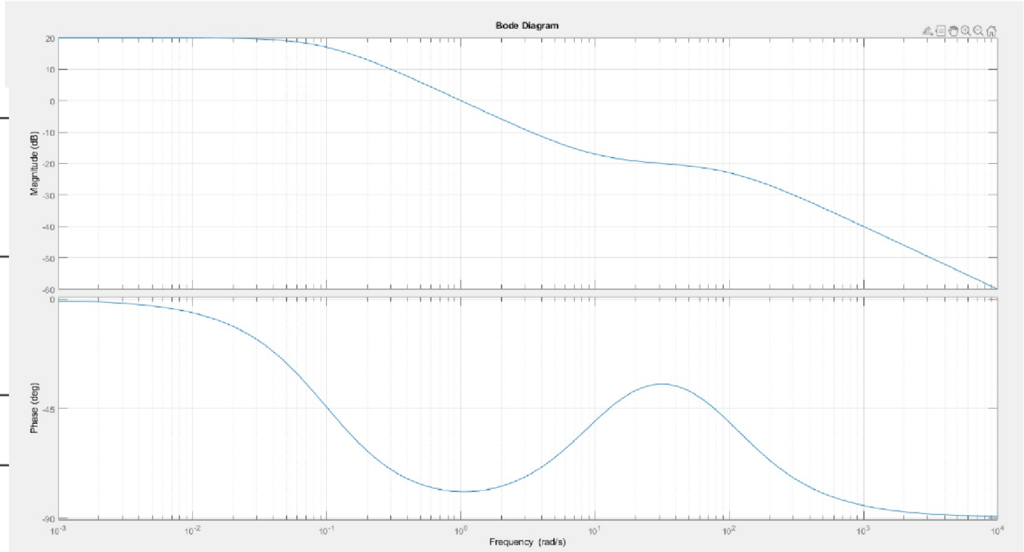
$$G(s) = 10\,\frac{(\frac{s}{10}+1)}{(\frac{s}{0.1}+1)(\frac{s}{-10}+1)}$$

Element type	Formally	Magnitude	Phase
single stable pole	$Re(p_i) \leq 0$	$-20\,\frac{dB}{dec}$	-90°
single unstable pole	$Re(p_i) > 0$	$-20\,\frac{dB}{dec}$	$+90^\circ$
cmplx stable pole	$Re(p_i) \leq 0$	$-40\,\frac{dB}{dec}$	-180°
cmplx unstable pole	$Re(p_i) > 0$	$-40\,\frac{dB}{dec}$	$+180^\circ$
single mnm phs zero	$Re(z_i) \leq 0$	$+20\,\frac{dB}{dec}$	$+90^\circ$
single non-mnm phs zero	$Re(z_i) > 0$	$+20\,\frac{dB}{dec}$	-90°
cmplx mnm phs zero	$Re(z_i) \leq 0$	$+40\,\frac{dB}{dec}$	$+180^\circ$
cmplx non-mnm phs zero	$Re(z_i) > 0$	$+40\,\frac{dB}{dec}$	-180°
positive constant	$k \geq 0$	$0\,\frac{dB}{dec}$	0°
negative constant	$k < 0$	$0\,\frac{dB}{dec}$	$\pm180^\circ$
differentiator	$\tau \cdot s$	$+20\,\frac{dB}{dec}$	$+90^\circ$
integrator	$\frac{1}{\tau \cdot s}$	$-20\,\frac{dB}{dec}$	-90°
time delay	$e^{s\cdot\tau}$	$0\,\frac{dB}{dec}$	$-\omega\cdot\tau$



Constructing TF from Bode Plot

Element type	Formally	Magnitude	Phase
single stable pole	$Re(p_i) \leq 0$	$-20\frac{dB}{dec}$	-90°
single unstable pole	$Re(p_i) > 0$	$-20\frac{dB}{dec}$	$+90^\circ$
cmplx stable pole	$Re(p_i) \leq 0$	$-40\frac{dB}{dec}$	-180°
cmplx unstable pole	$Re(p_i) > 0$	$-40\frac{dB}{dec}$	$+180^\circ$
single mnm phs zero	$Re(z_i) \leq 0$	$+20\frac{dB}{dec}$	$+90^\circ$
single non-mnm phs zero	$Re(z_i) > 0$	$+20\frac{dB}{dec}$	-90°
cmplx mnm phs zero	$Re(z_i) \leq 0$	$+40\frac{dB}{dec}$	$+180^\circ$
cmplx non-mnm phs zero	$Re(z_i) > 0$	$+40\frac{dB}{dec}$	-180°
positive constant	$k \geq 0$	$0\frac{dB}{dec}$	0°
negative constant	$k < 0$	$0\frac{dB}{dec}$	$\pm 180^\circ$
differentiator	$\tau \cdot s$	$+20\frac{dB}{dec}$	$+90^\circ$
integrator	$\frac{1}{\tau \cdot s}$	$-20\frac{dB}{dec}$	-90°
time delay	$e^{s \cdot \tau}$	$0\frac{dB}{dec}$	$-\omega \cdot \tau$



Old Example Problem

Problem: Given in Figure 9 are the Bode plots of four different transfer functions.

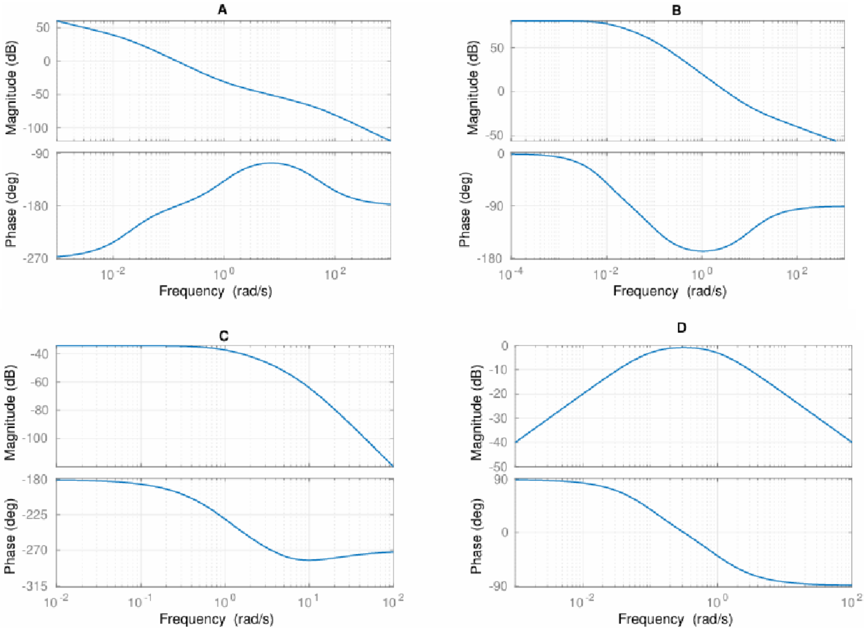


Figure 9: Bode plots of four different systems.

Q23 **(1.5 Points)** Assign each transfer function to the corresponding Bode plot.

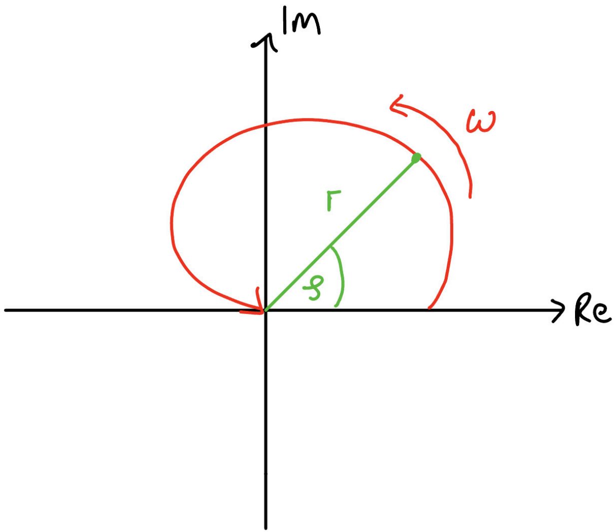
Transfer Function	A	B	C	D
$L_1(s) = \frac{s+1}{s(s+50)(s-0.02)}$				
$L_2(s) = \frac{s}{(s+1)(s+0.01)}$				
$L_3(s) = \frac{s+10}{(s+0.01)(s+0.1)}$				
$L_4(s) = \frac{1}{(s+1)(s-10)(s+5)}$				

Element type	Formally	Magnitude	Phase
single stable pole	$Re(p_i) \leq 0$	$-20\,\frac{dB}{dec}$	-90°
single unstable pole	$Re(p_i) > 0$	$-20\,\frac{dB}{dec}$	$+90^\circ$
cmplx stable pole	$Re(p_i) \leq 0$	$-40\,\frac{dB}{dec}$	-180°
cmplx unstable pole	$Re(p_i) > 0$	$-40\,\frac{dB}{dec}$	$+180^\circ$
single mnm phs zero	$Re(z_i) \leq 0$	$+20\,\frac{dB}{dec}$	$+90^\circ$
single non-mnm phs zero	$Re(z_i) > 0$	$+20\,\frac{dB}{dec}$	-90°
cmplx mnm phs zero	$Re(z_i) \leq 0$	$+40\,\frac{dB}{dec}$	$+180^\circ$
cmplx non-mnm phs zero	$Re(z_i) > 0$	$+40\,\frac{dB}{dec}$	-180°
positive constant	$k \geq 0$	$0\,\frac{dB}{dec}$	0°
negative constant	$k < 0$	$0\,\frac{dB}{dec}$	$\pm180^\circ$
differentiator	$\tau \cdot s$	$+20\,\frac{dB}{dec}$	$+90^\circ$
integrator	$\frac{1}{\tau \cdot s}$	$-20\,\frac{dB}{dec}$	-90°
time delay	$e^{s \cdot \tau}$	$0\,\frac{dB}{dec}$	$-\omega \cdot \tau$

3. Polar Plots

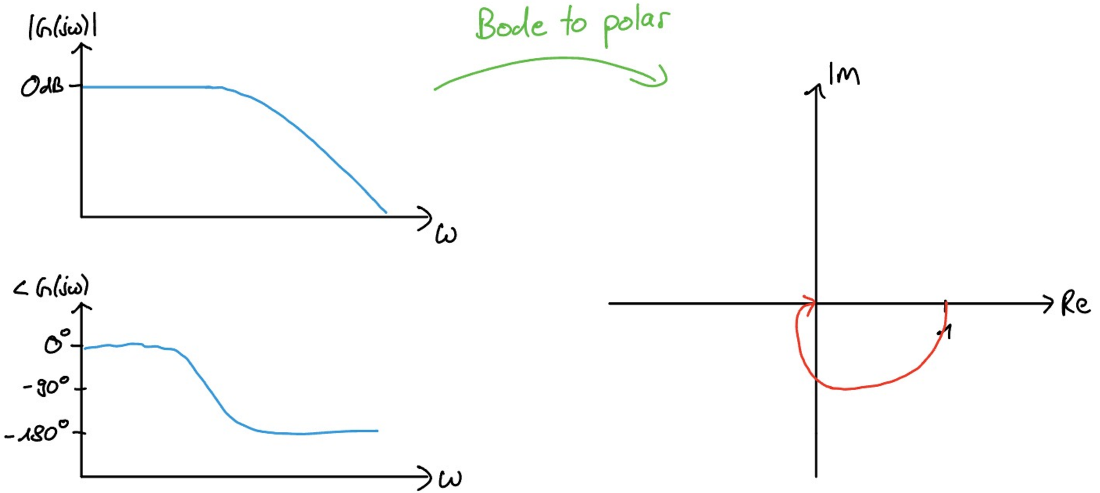
Polar Plot

- In this plot we combine magnitude & phase into a single complex number
- The frequency is no longer an axis but implicit, just like the gain k in Root Locus



Bode to Polar Plot

- If we have the Bode plot, we can extract the polar plot as such:



careful: 0dB = 1

Note: For all of these plots we are sketching, you won't have to precisely draw them, but have an intuition to correctly sketch them

I personally would recommend solving at least the problems marked red

Tips for Problem Set 8

Easy	Medium	Hard
	1, 2, 3, 4	5

1(i+ii): Make use of the Bode plot overview

1(iii+iv): Beware of the integrator and differentiator

2: Eliminate the easier ones first, and then the harder ones. Try looking at where the bode plot starts and what this means (integrator, differentiator, negative gain?)

3: Apply magnitude & phase rule, here I would rather try to understand the solutions

4: Look at the break points where the slopes change. Remember phase changes start one decade before and end one decade after in the phase plot. Doesn't have to be exact, if stuck consult solutions

5: Will help you understand nyquist plots better next week, but pretty hard imo, so work more with the solutions and try to comprehend it

Great visualization tool for Bode Plots from Johannes Schulte-Vels: https://bodetooltest.streamlit.app/

Questions?

Feedback?

Too fast? Too slow? Less theory, more exercises?

I would appreciate your feedback. Please let me know.

https://docs.google.com/forms/d/e/1FAIpQLSdHl0kjWo63aNzDkAV0cnmQadCAj5L0D7v7aSh0BK7BBdEgpA/viewform?usp=header